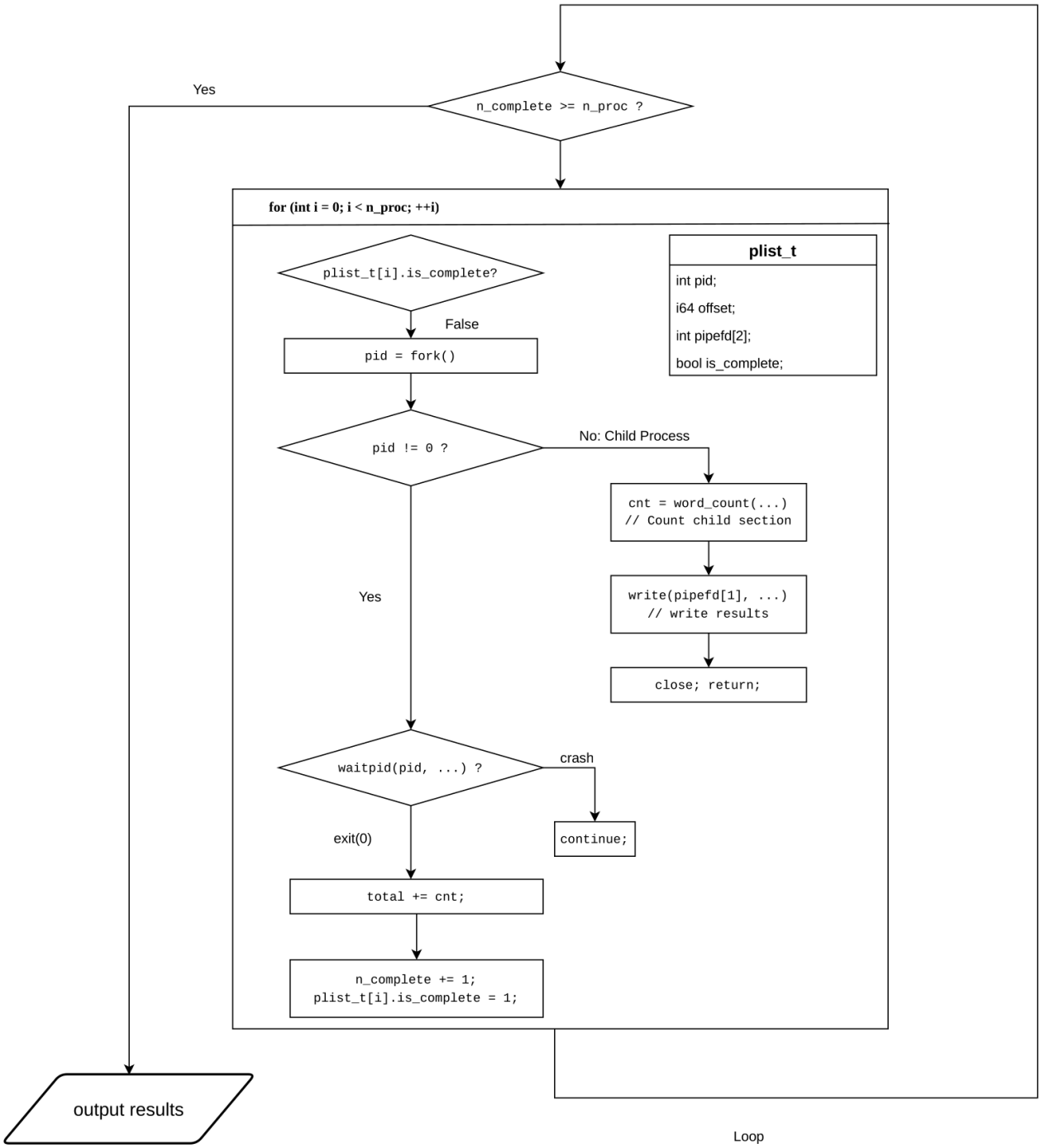
Kaiyuan Zhang; #811597925

Program Structure



IPC

Create pipe with

```
pipe(plist[i].pipefd);
```

In child process, write count information with :

```
write(plist[i].pipefd[1], &count, sizeof(count);
```

In parent process, read sub-section count information with:

```
read(plist[i].pipefd[0], &cnt, sizeof(count_t));
```

Handle Crashing

Keep variable `n_complete` to check if the task is completed, if `n_complete < n_process`, continue the loop.

If a child process crashes or exits abnormally, the parent detects the failure using `read()` byte-count validation and `waitpid()` status macros (`WIFEXITED` and `WEXITSTATUS`),
If crash:
`plist[i].is_complete == False`;
Else:
`plist[i].is_complete == True; ++n_complete`